PCE Equations

Final Prediction:

PCE = ((-0.1650*b_2_1) + (1.1259*spline(Path 1))) + ((0.4668*b_2_2) + (1.1868*spline(Path 2))) + ((-0.8068*b_2_3) + (0.9288*spline(Path 3))) + ((1.4720*b_2_4) + (1.0686*spline(Path 4))) + ((0.5658*b_2_5) + (1.5617*spline(Path 5))) + ((0.5874*b_2_6) + (1.6586*spline(Path 6))) + ((0.5512*b_2_7) + (1.2775*spline(Path 7))) + ((1.4698*b_2_8) + (1.8477*spline(Path 8))) + ((0.5034*b_2_9) + (1.9841*spline(Path 9))) + ((0.8613*b_2_10) + (1.3641*spline(Path 10)))

$$
\begin{aligned}
PCE = & ((-0.1650 * b_{2_1}) + (1.1259 * spline(Path1)))+ \\
& ((0.4668 * b_{2_2}) + (1.1868 * spline(Path2)))+ \\
& ((-0.8068 * b_{2_3}) + (0.9288 * spline(Path3)))+ \\
& ((1.4720 * b_{2_4}) + (1.0686 * spline(Path4)))+ \\
& ((0.5658 * b_{2_5}) + (1.5617 * spline(Path5)))+ \\
& ((0.5874 * b_{2_6}) + (1.6586 * spline(Path6)))+ \\
& ((0.5512 * b_{2_7}) + (1.2775 * spline(Path7)))+ \\
& ((1.4698 * b_{2_8}) + (1.8477 * spline(Path8)))+ \\
& ((0.5034 * b_{2_9}) + (1.9841 * spline(Path9)))+ \\
& ((0.8613 * b_{2_{10}}) + (1.3641 * spline(Path10)))
\end{aligned}
$$

Second Layer residuals:

b_{2_1} = \frac{Path1}{1+e^{-Path1}}
b_{2_2} = \frac{Path2}{1+e^{-Path2}}
b_{2_3} = \frac{Path3}{1+e^{-Path3}}
b_{2_4} = \frac{Path4}{1+e^{-Path4}}
b_{2_5} = \frac{Path5}{1+e^{-Path5}}
b_{2_6} = \frac{Path6}{1+e^{-Path6}}
b_{2_7} = \frac{Path7}{1+e^{-Path7}}
b_{2_8} = \frac{Path8}{1+e^{-Path8}}
b_{2_9} = \frac{Path9}{1+e^{-Path9}}
b_{2_{10}} = \frac{Path10}{1+e^{-Path10}}

$$b_{2_1} = \frac{Path1}{1+e^{-Path1}}$$
$$b_{2_2} = \frac{Path2}{1+e^{-Path2}}$$
$$b_{2_3} = \frac{Path3}{1+e^{-Path3}}$$
$$b_{2_4} = \frac{Path4}{1+e^{-Path4}}$$
$$b_{2_5} = \frac{Path5}{1+e^{-Path5}}$$
$$b_{2_6} = \frac{Path6}{1+e^{-Path6}}$$
$$b_{2_7} = \frac{Path7}{1+e^{-Path7}}$$
$$b_{2_8} = \frac{Path8}{1+e^{-Path8}}$$
$$b_{2_9} = \frac{Path9}{1+e^{-Path9}}$$
$$b_{2_{10}} = \frac{Path10}{1+e^{-Path10}}$$

First Layer Paths

Path 1 = ((-0.2282*b_1) + (0.2621*spline(PC1))) + (-.10225* tanh(-6.3724*PC1 - .2757) + .0458) + ((-0.0461*b_3) + (0.3115*spline(PC3))) + ((0.5109*b_4) + (0.3468*spline(PC4)))\\

Path 2 = (.0363 * \tanh(-9.9581*PC1 + 2.1827) - .0314) + ((0.2698*b_2) + (0.1794*spline(PC2))) + ((-0.3263*b_3) + (0.1876*spline(PC3))) + ((-0.1179*b_4) + (0.2365*spline(PC4)))\\

Path 3 = (-.0942* |(-5.9933*PC1 - 1.7912| + .2376) + ((-0.3595*b_2) + (0.2884*spline(PC2))) + ((0.0903*b_3) + (0.3872*spline(PC3))) + ((0.0596*b_4) + (0.3316*spline(PC4)))\\

Path 4 = ((0.1953*b_1) + (0.1347*spline(PC1))) + (-.0486 * \arctan(-9.5903*PC2 + .4677) + .0114) + ((0.1400*b_3) + (0.1427*spline(PC3))) + ((-0.1102*b_4) + (0.3556*spline(PC4)))\\

Path 5 = ((0.0427*b_1) + (0.5063*spline(PC1))) + ((0.1827*b_2) + (0.4409*spline(PC2))) + ((0.1038*b_3) + (0.5859*spline(PC3))) + ((-0.0235*b_4) + (0.4652*spline(PC4)))\\

Path 6 = (-.0306 * |-7.7262*PC1 - 2.1632| + .0200) + ((-0.0424*b_2) + (0.1441*spline(PC2))) + ((0.1831*b_3) + (0.3760*spline(PC3))) + ((-0.2691*b_4) + (0.2238*spline(PC4)))\\

Path 7 = (-.3629*gaussian(-2.3925*PC1 - .5070) + .3666) + ((0.6870*b_2) + (0.5736*spline(PC2))) + ((0.4406*b_3) + (0.5150*spline(PC3))) + ((0.1439*b_4) + (0.4731*spline(PC4)))\\

Path 8 = ((-0.0730*b_1) + (0.6707*spline(PC1))) + ((0.1651*b_2) + (0.5811*spline(PC2))) + ((0.0934*b_3) + (0.5168*spline(PC3))) + ((-0.1729*b_4) + (0.5478*spline(PC4)))\\

Path 9 = ((-0.0847*b_1) + (0.4732*spline(PC1))) + ((-0.1634*b_2) + (0.5161*spline(PC2))) + ((-0.0256*b_3) + (0.4321*spline(PC3))) + ((0.0142*b_4) + (0.4819*spline(PC4)))\\

Path 10 = ((-0.2318*b_1) + (0.2480*spline(PC1))) + ((0.0978*b_2) + (0.2692*spline(PC2))) + ((0.2514*b_3) + (0.3138*spline(PC3))) + ((-0.0501*b_4) + (0.3104*spline(PC4))\\

$$Path1 = ((-0.2282 * b_1) + (0.2621 * spline(PC1))) + (-.10225 * tanh(-6.3724 * PC1 - .2757) + .0458) + ((-0.0461 * b_3) + (0.3115 * spline(PC3))) + ((0.5109 * b_4) + (0.3468 * spline(PC4)))$$

$$Path2 = (.0363 * \tanh(-9.9581 * PC1 + 2.1827) - .0314) + ((0.2698 * b_2) + (0.1794 * spline(PC2))) + ((-0.3263 * b_3) + (0.1876 * spline(PC3))) + ((-0.1179 * b_4) + (0.2365 * spline(PC4)))$$

$$Path3 = (-.0942 * |(-5.9933 * PC1 - 1.7912| + .2376) + ((-0.3595 * b_2) + (0.2884 * spline(PC2))) + ((0.0903 * b_3) + (0.3872 * spline(PC3))) + ((0.0596 * b_4) + (0.3316 * spline(PC4)))$$

$$Path4 = ((0.1953 * b_1) + (0.1347 * spline(PC1))) + (-.0486 * \arctan(-9.5903 * PC2 + .4677) + .0114) + ((0.1400 * b_3) + (0.1427 * spline(PC3))) + ((-0.1102 * b_4) + (0.3556 * spline(PC4)))$$

$$Path5 = ((0.0427 * b_1) + (0.5063 * spline(PC1))) + ((0.1827 * b_2) + (0.4409 * spline(PC2))) + ((0.1038 * b_3) + (0.5859 * spline(PC3))) + ((-0.0235 * b_4) + (0.4652 * spline(PC4)))$$

$$Path6 = (-.0306 * | - 7.7262 * PC1 - 2.1632| + .0200) + ((-0.0424 * b_2) + (0.1441 * spline(PC2))) + ((0.1831 * b_3) + (0.3760 * spline(PC3))) + ((-0.2691 * b_4) + (0.2238 * spline(PC4)))$$

$$Path7 = (-.3629 * gaussian(-2.3925 * PC1 - .5070) + .3666) + ((0.6870 * b_2) + (0.5736 * spline(PC2))) + ((0.4406 * b_3) + (0.5150 * spline(PC3))) + ((0.1439 * b_4) + (0.4731 * spline(PC4)))$$

$$Path8 = ((-0.0730 * b_1) + (0.6707 * spline(PC1))) + ((0.1651 * b_2) + (0.5811 * spline(PC2))) + ((0.0934 * b_3) + (0.5168 * spline(PC3))) + ((-0.1729 * b_4) + (0.5478 * spline(PC4)))$$

$$Path9 = ((-0.0847 * b_1) + (0.4732 * spline(PC1))) + ((-0.1634 * b_2) + (0.5161 * spline(PC2))) + ((-0.0256 * b_3) + (0.4321 * spline(PC3))) + ((0.0142 * b_4) + (0.4819 * spline(PC4)))$$

$$Path10 = ((-0.2318 * b_1) + (0.2480 * spline(PC1))) + ((0.0978 * b_2) + (0.2692 * spline(PC2))) + ((0.2514 * b_3) + (0.3138 * spline(PC3))) + ((-0.0501 * b_4) + (0.3104 * spline(PC4))$$

<u>Residual Equations:</u>

b_1 = \frac{PC1}{1 + e^{-PC1}}
b_2 = \frac{PC2}{1 + e^{-PC2}}
b_3 = \frac{PC3}{1 + e^{-PC3}}
b_4 = \frac{PC4}{1 + e^{-PC4}}

$$b_1 = \frac{PC1}{1 + e^{-PC1}}$$

$$b_2 = \frac{PC2}{1 + e^{-PC2}}$$

$$b_3 = \frac{PC3}{1 + e^{-PC3}}$$

$$b_4 = \frac{PC4}{1 + e^{-PC4}}$$

<u>PCA Equations</u>

PC1 = 0.346*S_{homo}(HOMO) + -0.286*S_{lumo}(LUMO) + -0.355*S_{gap}(Gap) + 0.415*S_{single}(Single) + 0.616*S_{double}(Double) + 0.335*S_{triple}(Triple) + 0.090*S_{aromatic}(Aromatic)
PC2 = 0.600*S_{homo}(HOMO) + -0.232*S_{lumo}(LUMO) + -0.439*S_{gap}(Gap) + -0.180*S_{single}(Single) + -0.485*S_{double}(Double) + -0.239*S_{triple}(Triple) + 0.263*S_{aromatic}(Aromatic)
PC3 = -0.179*S_{homo}(HOMO) + 0.119*S_{lumo}(LUMO) + 0.164*S_{gap}(Gap) + 0.504*S_{single}(Single) + -0.287*S_{double}(Double) + 0.160*S_{triple}(Triple) + 0.751*S_{aromatic}(Aromatic)
PC4 = -0.016*S_{homo}(HOMO) + -0.090*S_{lumo}(LUMO) + -0.053*S_{gap}(Gap) + -0.210*S_{single}(Single) + -0.376*S_{double}(Double) + 0.881*S_{triple}(Triple) + -0.167*S_{aromatic}(Aromatic)

$$PC1 = 0.346*S_{homo}(HOMO) + -0.286*S_{lumo}(LUMO) + -0.355*S_{gap}(Gap) + 0.415*S_{single}(Single) + 0.616*S_{double}(Double) + 0.335*S_{triple}(Triple) + 0.090*S_{aromatic}(Aromatic)$$
$$PC2 = 0.600*S_{homo}(HOMO) + -0.232*S_{lumo}(LUMO) + -0.439*S_{gap}(Gap) + -0.180*S_{single}(Single) + -0.485*S_{double}(Double) + -0.239*S_{triple}(Triple) + 0.263*S_{aromatic}(Aromatic)$$
$$PC3 = -0.179*S_{homo}(HOMO) + 0.119*S_{lumo}(LUMO) + 0.164*S_{gap}(Gap) + 0.504*S_{single}(Single) + -0.287*S_{double}(Double) + 0.160*S_{triple}(Triple) + 0.751*S_{aromatic}(Aromatic)$$
$$PC4 = -0.016*S_{homo}(HOMO) + -0.090*S_{lumo}(LUMO) + -0.053*S_{gap}(Gap) + -0.210*S_{single}(Single) + -0.376*S_{double}(Double) + 0.881*S_{triple}(Triple) + -0.167*S_{aromatic}(Aromatic)$$

Scaling Equations:

S_{homo}(x) = \frac{x + 0.280}{0.125}
S_{lumo}(x) = \frac{x + 0.163}{0.178}
S_{gap}(x) = \frac{x - 0.029}{0.265}
S_{single}(x) = \frac{x - 7.000}{76}
S_{double}(x) = \frac{x}{16}
S_{triple}(x) = \frac{x}{4}
S_{aromatic}(x) = \frac{x}{70}

$$S_{homo}(x) = \frac{x + 0.280}{0.125}$$
$$S_{lumo}(x) = \frac{x + 0.163}{0.178}$$
$$S_{gap}(x) = \frac{x - 0.029}{0.265}$$
$$S_{single}(x) = \frac{x - 7.000}{76}$$
$$S_{double}(x) = \frac{x}{16}$$
$$S_{triple}(x) = \frac{x}{4}$$
$$S_{aromatic}(x) = \frac{x}{70}$$

Complement Equations

Final Predictions:

Logit C60 = .754*gaussian(-.388*Path1 + 1.665) - .313 + .833*tan(-9.953 * (Path 2) - 2.818) - .164 + ((0.082*b_2_3) + (3.405*spline(Path 3))) + arctan(-6.177*Path 4 + 3.070 + -2.1542e+00 + 5.8716e-01) + -.542 * gaussian(-.477*Path 5 + 1.548) + .005

Logit PC61BM = ((-1.468*b_2_1) + (3.584*spline(Path 1))) + ((0.828*b_2_2) + (6.877*spline(Path 2))) + ((0.325*b_2_3) + (4.450*spline(Path 3))) + ((-3.559*b_2_4) + (3.426*spline(Path 4))) + (-158.34/(-3.032*Path 5 + 9.480)^2) + 3.003

Logit PC71BM = ((4.199*b_2_1) + (5.175*spline(Path 1))) + ((1.631*b_2_2) + (0.704*spline(Path 2))) + ((-2.244*b_2_3) + (2.279*spline(Path 3))) + ((1.419*b_2_4) + (5.953*spline(Path 4))) + ((-0.085*b_2_5) + (-5.098*spline(Path 5)))

Logit TiO2 = ((-0.313*b_2_1) + (1.792*spline(Path 1))) + ((-1.017*b_2_2) + (1.299*spline(Path 2))) + ((1.884*b_2_3) + (2.361*spline(Path 3))) + ((-1.147*b_2_4) + (2.734*spline(Path 4))) + -3.163*tanh(-1.147*Path 5 + 3.033) + 1.155

$$\text{Logit C60} = .754{*}\text{gaussian}(-.388{*}\text{Path1} + 1.665) - .313 + .833{*}\tan(-9.953 * (\text{Path 2}) - 2.818) - .164 + ((0.082{*}\text{b}_{2_3}) + (3.405*spline(Path3))) + -2.154*arctan(-6.177*Path4+3.070)+.587+-.542*gaussian(-.477*Path5+1.548)+.005$$

$$\text{Logit PC61BM} = ((\text{-}1.468{*}\text{b}_{2_1}) + (3.584*spline(Path1))) + ((0.828*b_{2_2}) + (6.877*spline(Path2))) + ((0.325*b_{2_3}) + (4.450*spline(Path3))) + ((-3.559*b_{2_4}) + (3.426*spline(Path4))) + (-158.34/(-3.032*Path5+9.480)^2) + 3.003$$

$$\text{Logit PC71BM} = ((4.199{*}\text{b}_{2_1}) + (5.175*spline(Path1))) + ((1.631*b_{2_2}) + (0.704*spline(Path2))) + ((-2.244*b_{2_3}) + (2.279*spline(Path3))) + ((1.419*b_{2_4}) + (5.953*spline(Path4))) + ((-0.085*b_{2_5}) + (-5.098*spline(Path5)))$$

$$\text{Logit TiO2} = ((\text{-}0.313{*}\text{b}_{2_1}) + (1.792*spline(Path1))) + ((-1.017*b_{2_2}) + (1.299*spline(Path2))) + ((1.884*b_{2_3}) + (2.361*spline(Path3))) + ((-1.147*b_{2_4}) + (2.734*spline(Path4))) + -3.163*tanh(-1.147*Path5+3.033)+1.155$$

Second Layer Residuals:

b_2_1 = \frac{Path 1}{1 + e^{-Path 1}}
b_2_2 = \frac{Path 2}{1 + e^{-Path 2}}
b_2_3 = \frac{Path 3}{1 + e^{-Path 3}}
b_2_4 = \frac{Path 4}{1 + e^{-Path 4}}

b_2_5 = \frac{Path 5}{1 + e^{-Path 5}}

$$b_{2_1} = \frac{Path1}{1 + e^{-Path1}}$$

$$b_{2_2} = \frac{Path2}{1 + e^{-Path2}}$$

$$b_{2_3} = \frac{Path3}{1 + e^{-Path3}}$$

$$b_{2_4} = \frac{Path4}{1 + e^{-Path4}}$$

$$b_{2_5} = \frac{Path5}{1 + e^{-Path5}}$$

First Layer Paths

Path 1 = -1.240*tanh(-0.841*PC1 + 1.3013) + 1.409 + -2.681*gaussian(-0.321*PC2 + 0.993) + 0.597 + -3.215*sin(-0.427 + 7.294) + 2.795 + ((0.435*b_4) + (0.666*spline(PC4)))

Path 2 = ((0.104*b_1) + (0.146*spline(PC1))) + -0.031*|-5.871*PC2 - 5.433| + 0.214 + 15.271*cos(-0.073*PC3 - 6.569) - 14.848 + ((-0.385*b_4) + (0.469*spline(PC4)))

Path 3 = 7.644*tanh(-0.500*PC1 + 1.122) - 6.670 + -0.925*gaussian(-0.6715*PC2 - 0.7004) + 0.4754 + 2.019*tanh(-0.6245*PC3 + 1.544) - 1.644 + 2.269*tanh(-0.932*PC4 + 1.871) - 2.495

Path 4 = 0.942*tanh(-0.5677*PC1 + 1.236) - 0.614 + -1.712*cos(-0.460*PC2 - 3.167) - 1.723 + ((-0.322*b_3) + (0.534*spline(PC3))) + ((-0.214*b_4) + (0.646*spline(PC4)))

Path 5 = ((-0.259*b_1) + (0.639*spline(PC1))) + ((0.373*b_2) + (0.496*spline(PC2))) + -17.912*gaussian(-0.318*PC3 + 1.441) + 1.719 + ((-1.226*b_4) + (1.886*spline(PC4)))

Path 1 = -1.240*tanh(-0.841*PC1 + 1.3013) + 1.409 + -2.681*gaussian(-0.321*PC2 + 0.993) + 0.597 + -3.215*sin(-0.427 + 7.294) + 2.795 + ((0.435*b_4)+ $(0.666 * spline(PC4)))$

Path 2 = ((0.104*b_1)+$(0.146*spline(PC1)))+-0.031*|-5.871*PC2-5.433|+0.214+15.271*cos(-0.073*PC3-6.569)-14.848+((-0.385*b_4)+(0.469*spline(PC4)))$

Path 3 = 7.644*tanh(-0.500*PC1 + 1.122) - 6.670 + -0.925*gaussian(-0.6715*PC2 - 0.7004) + 0.4754 + 2.019*tanh(-0.6245*PC3 + 1.544) - 1.644 + 2.269*tanh(-0.932*PC4 + 1.871) - 2.495

Path 4 = 0.942*tanh(-0.5677*PC1 + 1.236) - 0.614 + -1.712*cos(-0.460*PC2 - 3.167) - 1.723 + ((-0.322*b_3) + $(0.534 * spline(PC3))) + ((-0.214 * b_4) + (0.646 * spline(PC4)))$

Path 5 = ((-0.259*b_1) + $(0.639 * spline(PC1))) + ((0.373 * b_2) + (0.496 * spline(PC2)))+-17.912*gaussian(-0.318*PC3+1.441)+1.719+((-1.226*b_4)+(1.886*spline(PC4)))$

Residual Equations:

b_1 = \frac{PC1}{1 + e^{-PC1}}
b_2 = \frac{PC2}{1 + e^{-PC2}}
b_3 = \frac{PC3}{1 + e^{-PC3}}
b_4 = \frac{PC4}{1 + e^{-PC4}}

$$b_1 = \frac{PC1}{1 + e^{-PC1}}$$

$$b_2 = \frac{PC2}{1 + e^{-PC2}}$$

$$b_3 = \frac{PC3}{1 + e^{-PC3}}$$

$$b_4 = \frac{PC4}{1 + e^{-PC4}}$$

PCA Equations

PC1 = -0.513*S_{homo}(HOMO) + 0.542*S_{lumo}(LUMO) + 0.582*S_{gap}(Gap) + -0.197*S_{single}(Single) + -0.157*S_{double}(Double) + -0.142*S_{triple}(Triple) + -0.142*S_{aromatic}(Aromatic)
PC2 = -0.289*S_{homo}(HOMO) + 0.032*S_{lumo}(LUMO) + 0.184*S_{gap}(Gap) + 0.554*S_{single}(Single) + 0.543*S_{double}(Double) + 0.529*S_{triple}(Triple) + 0.020*S_{aromatic}(Aromatic)
PC3 = 0.010*S_{homo}(HOMO) + 0.155*S_{lumo}(LUMO) + 0.076*S_{gap}(Gap) + 0.352*S_{single}(Single) + -0.368*S_{double}(Double) + -0.053*S_{triple}(Triple) + 0.841*S_{aromatic}(Aromatic)
PC4 = -0.012*S_{homo}(HOMO) + -0.036*S_{lumo}(LUMO) + -0.012*S_{gap}(Gap) + -0.351*S_{single}(Single) + -0.444*S_{double}(Double) + 0.823*S_{triple}(Triple) + 0.012*S_{aromatic}(Aromatic)

$$PC1 = -0.513 * S_{homo}(HOMO) + 0.542 * S_{lumo}(LUMO) + 0.582 * S_{gap}(Gap) + -0.197 * S_{single}(Single) + -0.157 * S_{double}(Double) + -0.142 * S_{triple}(Triple) + -0.142 * S_{aromatic}(Aromatic)$$

$$PC2 = -0.289 * S_{homo}(HOMO) + 0.032 * S_{lumo}(LUMO) + 0.184 * S_{gap}(Gap) + 0.554 * S_{single}(Single) + 0.543 * S_{double}(Double) + 0.529 * S_{triple}(Triple) + 0.020 * S_{aromatic}(Aromatic)$$

$$PC3 = 0.010 * S_{homo}(HOMO) + 0.155 * S_{lumo}(LUMO) + 0.076 * S_{gap}(Gap) + 0.352 * S_{single}(Single) + -0.368 * S_{double}(Double) + -0.053 * S_{triple}(Triple) + 0.841 * S_{aromatic}(Aromatic)$$

$$PC4 = -0.012 * S_{homo}(HOMO) + -0.036 * S_{lumo}(LUMO) + -0.012 * S_{gap}(Gap) + -0.351 * S_{single}(Single) + -0.444 * S_{double}(Double) + 0.823 * S_{triple}(Triple) + 0.012 * S_{aromatic}(Aromatic)$$

<u>Scaling Equations:</u>

S_{homo}(x) = \frac{x + 0.196}{0.022}
S_{lumo}(x) = \frac{x + 0.099}{0.020}
S_{gap}(x) = \frac{x - 0.097}{0.038}
S_{single}(x) = \frac{x - 37.110}{13.118}
S_{double}(x) = \frac{x - 2.071}{3.438}
S_{triple}(x) = \frac{x - 0.214}{0.670}
S_{aromatic}(x) = \frac{x - 35.287}{11.995}

$$S_{homo}(x) = \frac{x + 0.196}{0.022}$$

$$S_{lumo}(x) = \frac{x + 0.099}{0.020}$$

$$S_{gap}(x) = \frac{x - 0.097}{0.038}$$

$$S_{single}(x) = \frac{x - 37.110}{13.118}$$

$$S_{double}(x) = \frac{x - 2.071}{3.438}$$

$$S_{triple}(x) = \frac{x - 0.214}{0.670}$$

$$S_{aromatic}(x) = \frac{x - 35.287}{11.995}$$

Gaussian definition:
$Y = e^{-x^2}$